George Dare-Williams

Aspiring Software Engineer & Systems Admin

London | g.darewilliams@gmail.com | big-george.net | github.com/thatbluebean

Personal Summary

I am a software and systems enthusiast with a passion for open-source technology, as well as a 3D printing and robotics hobbyist. I write efficient Python and Bash scripts, with experience in C#, JS and Rust. A tinkerer at heart, I enjoy exploring new technologies, and work on my personal linux homelab in my free time.

From custom Linux environments to containerised workloads, I focus on performance, automation and **reliable infrastructure design**, and I am always **keen to learn** new technologies and concepts, as well as how I can deploy them in real world scenarios.

Industry Work Experience

Developer / Project Leader July 2026

Digital Catapult, London

- Project manager on a work experience programme, secured through a competitive process of structured applications and online interviews.
- Worked with **peers and employees** to develop a pitch for a mock startup focusing on improving public transport through future 'deep tech' integrations, **pitched to real investors**.

Robotics Developer July 2026

University College London (UCL), Stratford campus

- Robotics research placement with competitive application process.
- Built computer vision models and kinematics to let users control a robot arm with their own arm movements.
- Worked **with a team of university students** to build and maintain a GitHub Python codebase, learning a range of new libraries.
- **My main contribution was the computer vision pipeline**, translating camera inputs into commands sent to the robot arm. I coordinated with other team members working on the AI brain for the opponent.

Data Analyst / Developer February 2026

Bank of America, Central London

- One-week placement working in Python with Pandas **alongside developers, market analysts and economists** to produce and present stock market trend and prediction analysis, based on real data.
- **Maintained a codebase with Git**, gave and received feedback within the team, and delivered a finished product.

Software Engineer October 2024

BAE Systems, Rochester

- One-week placement working primarily in C++ alongside engineers who designed military night vision and fighter jet navigation systems.
- Maintained a codebase within a team of experienced programmers using Git and the GitHub CLI, **learning the basics of real libraries used at BAE.**

Employment

Stock & Sales Associate June 2025 – Present

Mobile Pet Foods Ltd

- Managing physical stock within a small, coordinated team.
- Serving customers directly, handling specialised orders as well as over-the-counter sales.

Technical Skills

Languages: Python, Bash scripting, HTML, JavaScript, C++, Rust, C#

DevOps & Servers: Docker, Linux (Proxmox, Arch, Ubuntu, Gentoo), Nginx, Git / GitHub Actions

Infrastructure & Homelab: Self-hosting, Proxmox VE, reverse proxies, local LLMs, Ubuntu/Debian/Nix, Tailscale/WireGuard VPN

Other: OpenCV, Pandas, CAD

Selected Projects

ESP32 Remote Message System | github.com/thatbluebean/ESP32-Messenger

- Built a system for sending text to an ESP32-powered LED matrix display from anywhere on the network.
- Developed a Flask HTTP server on a VPS which receives messages as JSON for the ESP32 to retrieve and show in real time.
- Designed in CAD and 3D printed a custom enclosure to make the standalone device user friendly.

Tech: Python, C++, HTML, CAD & 3D printing

Proxmox Homelab

- Custom-built homelab for deploying and testing applications and self-hosting services including Home Assistant, Jellyfin and Proxmox Backup, plus local backups of all software projects.
- Hardware: two mini PCs, a Raspberry Pi and a small NAS, housed in a 3D printed enclosure adapted from public designs to fit the hardware.
- Remote access via Tailscale with, with the Proxmox node acting as a subnet router for services.
- Interacts with my VPSs for certain tasks.

Tech: Docker, Proxmox/Linux, CAD, sysadmin

ShadowChat – TUI Chatroom | github.com/thatbluebean/shadowchat

- Joint project with a fellow student: a LAN messaging chatroom for the terminal
- Widely used by students at school, with binaries for any operating system.

Tech: Python

big-george.net | github.com/thatbluebean/big-george.net

- Self-hosted portfolio site on an Oracle Cloud VPS served by Nginx; a Bash script on the VPS pulls repo updates and reloads the web server configuration.

Tech: HTML, CSS, Bash, server administration

Education

Ada, the National College for Digital Skills London September 2025 - Present

- A-level Mathematics predicted A
- A-level Further Mathematics predicted A
- BTEC National Diploma Computing predicted Distinction* Distinction* (Equivalent to A*A*)

Bexley Grammar School September 2020 - July 2025

- GCSE Mathematics 8
- GCSE English Language 8
- GCSE English Literature 6
- GCSE Computer Science 8
- GCSE Drama 8
- L2 Further Mathematics 7
- GCSE Chemistry 6
- GCSE Physics 6
- GCSE Biology 6
- GCSE History 6

Interests

- Films, Building smart home components, Gaming, Linux, Open source contributions, Swimming, Gym.